PYTHON FOR DATA SCIENCE · LESSON 8 OF 29

# Dictionaries: Key-Value Data, Counting and Grouping

Store and look up data by key. Create and update dictionaries, use get, setdefault, items, Counter and defaultdict, nest dictionaries, and convert between dictionaries, JSON-like records and DataFrames using the Superstore data.

Python class 6 | Python 3.13 | Beginner | 100 min

**By the end of this lesson you can**

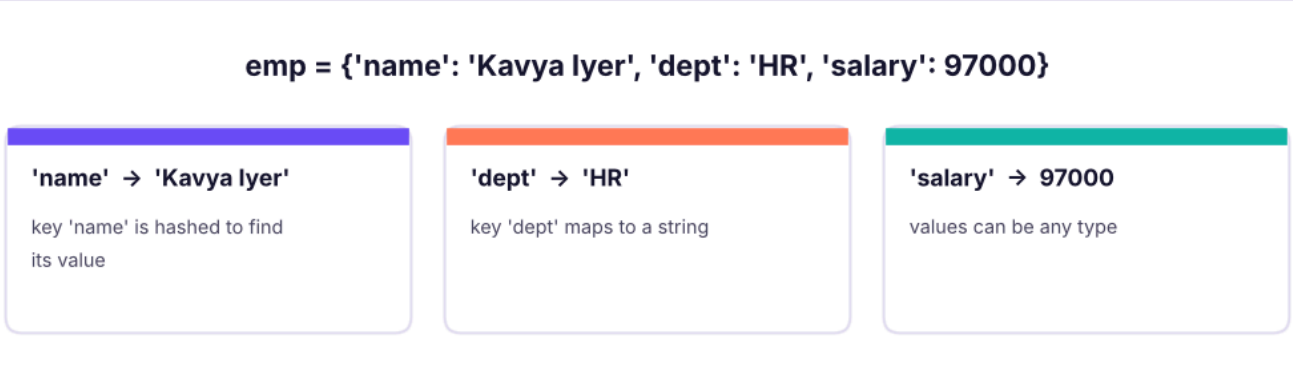
- Create dictionaries and read, add, update and delete keys safely
- Loop over keys, values and items and merge dictionaries
- Count and group with get, setdefault, Counter and defaultdict
- Work with nested dictionaries and lists of dictionaries (JSON-like records)
- Convert between dictionaries and pandas DataFrames

**Datasets used**

`superstore_sales.csv`

## What is a dictionary?

A **dictionary** ( `dict` ) maps **keys** to **values**: look up by name instead of by position. It keeps insertion order (Python 3.7+), and lookups are O(1) on average.




A dictionary is a set of key → value pairs. The key is hashed, so the lookup does not scan the data.

Python

```
emp = {"name": "Kavya Iyer", "dept": "HR", "salary": 97000}
print(emp["name"], emp.get("dept"), emp.get("city"), emp.get("city", "unknown"), len(emp),
"salary" in emp)
try: emp["city"]
except KeyError as e: print("KeyError:", e)
```

OUTPUT

```
Kavya Iyer HR None unknown 3 True
KeyError: 'city'
```

`d[key]` raises `KeyError` when missing; `d.get(key, default)` never does.

## Changing a dictionary

Python

```
emp["city"] = "Pune"                # add
emp["salary"] += 3000                # update
emp.update({"dept": "Finance", "grade": "M2"})
removed = emp.pop("grade")          # remove and return
del emp["city"]
print(emp, removed)
print(emp.setdefault("level", "L3"), emp.setdefault("dept", "X"), emp)    # set only if
missing
```

OUTPUT

```
{'name': 'Kavya Iyer', 'dept': 'Finance', 'salary': 100000} M2
L3 Finance {'name': 'Kavya Iyer', 'dept': 'Finance', 'salary': 100000, 'level': 'L3'}
```

| Method | Effect |
| --- | --- |
| `d[k] = v` | Add or overwrite |
| `d.get(k, default)` | Safe read |
| `d.setdefault(k, default)` | Read; insert default if missing |
| `d.update(other)` or `d \| other` | Merge (right side wins) |
| `d.pop(k[, default])` | Remove and return |
| `d.popitem()` | Remove the last pair |
| `del d[k]` | Delete (KeyError if missing) |
| `d.keys()`, `d.values()`, `d.items()` | Views |
| `d.copy()` | Shallow copy |
| `dict.fromkeys(keys, value)` | Create with the same value |

## Looping

Python

```python
prices = {"Pen": 10, "Notebook": 45, "Stapler": 180}
for k in prices: print(k, end=" ")
print()
for k, v in prices.items(): print(f"{k:<10}{v:>5}")
print(list(prices.values()), sum(prices.values()), max(prices, key=prices.get),
sorted(prices.items(), key=lambda kv: -kv[1]))
merged = prices | {"Pen": 12, "Folder": 30}              # Python 3.9+
print(merged)
print({v: k for k, v in prices.items()})                  # invert (values must be unique
and hashable)
```

OUTPUT

```
Pen Notebook Stapler
Pen          10
Notebook     45
Stapler     180
[10, 45, 180] 235 Stapler [('Stapler', 180), ('Notebook', 45), ('Pen', 10)]
{'Pen': 12, 'Notebook': 45, 'Stapler': 180, 'Folder': 30}
{10: 'Pen', 45: 'Notebook', 180: 'Stapler'}
```

# Counting and grouping

The most common dictionary jobs are **count** and **group**:

Python

```
from collections import Counter, defaultdict
words = "the cat and the hat and the bat".split()
manual = {}
for w in words: manual[w] = manual.get(w, 0) + 1            # classic pattern
print(manual)
c = Counter(words)
print(c, c.most_common(2), c["the"], c["dog"])
groups = defaultdict(list)                                   # missing key starts as an
empty list
for w in words: groups[len(w)].append(w)
print(dict(groups))
```

OUTPUT

```
{'the': 3, 'cat': 1, 'and': 2, 'hat': 1, 'bat': 1}
Counter({'the': 3, 'and': 2, 'cat': 1, 'hat': 1, 'bat': 1}) [('the', 3), ('and', 2)]
3 0
{3: ['the', 'cat', 'and', 'the', 'hat', 'and', 'the', 'bat']}
```

Python

```
import pandas as pd
s = pd.read_csv("superstore_sales.csv")
by_region = Counter(s["region"])                             # count rows
revenue = defaultdict(float)
for r, amount in zip(s["region"], s["sales"]): revenue[r] += amount
print(by_region.most_common())
print({k: round(v / 1e6, 1) for k, v in sorted(revenue.items(), key=lambda kv: -kv[1])}, "
(sales in millions)")
print(s.groupby("region")
["sales"].sum().div(1e6).round(1).sort_values(ascending=False).to_dict())   # same in
pandas
```

OUTPUT

```
[('South', 1656), ('West', 1614), ('North', 1530), ('East', 1523)]
{'South': 54.0, 'West': 52.0, 'North': 49.0, 'East': 46.5} (sales in millions)
{'South': 54.0, 'West': 52.0, 'North': 49.0, 'East': 46.5}
```

## Nested dictionaries and JSON-like records

Real data (API responses, configs, documents) nests dictionaries and lists. Python's `json` module reads and writes exactly this structure.

Python

```
import json
order = {"order_id": "O-1001", "customer": {"id": "C7", "name": "Asha", "city": "Pune"},
         "items": [{"sku": "P1", "qty": 2, "price": 450.0}, {"sku": "P2", "qty": 1,
"price": 1200.0}]}
total = sum(i["qty"] * i["price"] for i in order["items"])
print(order["customer"]["city"], len(order["items"]), total, order.get("shipping",
{}).get("city", "n/a"))
text = json.dumps(order, indent=2)
print(text[:90].replace("\n", " "), "...", json.loads(text) == order)
```

OUTPUT

```
Pune 2 2100.0 n/a
{   "order_id": "O-1001",   "customer": {     "id": "C7",     "name": "Asha",
"city":  ... True
```

A **list of dictionaries** is the standard row format, and it converts straight to a DataFrame:

Python

```
rows = [{"sku": "P1", "qty": 2, "price": 450.0}, {"sku": "P2", "qty": 1, "price": 1200.0},
{"sku": "P3", "qty": 5, "price": 80.0}]
df = pd.DataFrame(rows)
df["value"] = df.qty * df.price
print(df)
print(df.set_index("sku")["value"].to_dict(), df.to_dict("records")[0])
```

OUTPUT

```
  sku  qty   price   value
0  P1    2   450.0   900.0
1  P2    1  1200.0  1200.0
2  P3    5    80.0   400.0
{'P1': 900.0, 'P2': 1200.0, 'P3': 400.0} {'sku': 'P1', 'qty': 2, 'price': 450.0,
'value': 900.0}
```

## Lookup tables and dispatch

Python

```
tier = {"North": "A", "South": "A", "East": "B", "West": "C"}                #
mapping/lookup
print(s["region"].map(tier).value_counts().to_dict())
ops = {"add": lambda a, b: a + b, "mul": lambda a, b: a * b}                  # dictionary
of functions
print(ops["add"](3, 4), ops["mul"](3, 4))
```

OUTPUT

```
{'A': 3186, 'C': 1614, 'B': 1523}
7 12
```

NOTE **Keys must be hashable**

Strings, numbers and tuples work as keys; lists, sets and dicts do not. Use a tuple `("North", 2025)` for composite keys. Copying: `d.copy()` is shallow; use `copy.deepcopy` for nested data (Lesson 6).

## Common mistakes

| Mistake | Fix |
|---|---|
| `d["missing"]` raises KeyError | `d.get(k, default)` or `if k in d` |
| Changing a dictionary's size while looping over it | Loop over `list(d)` or build a new dict |
| `{}` for an empty set | `set()`; `{}` is an empty dict |
| Duplicate keys in a literal | The last one silently wins |
| Assuming a shallow copy protects nested data | `copy.deepcopy` |

# Summary

**KEY TAKEAWAYS**

- A dict maps hashable keys to values with O(1) average lookup and keeps insertion order.
- Use `get` for safe reads, `setdefault` / `defaultdict` for grouping, `Counter` for counting, `d | other` to merge.
- Loop with `items()`; build with dict comprehensions `{k: f(v) for k, v in ...}`.
- Nested dict/list structures are JSON; a list of dicts converts to a DataFrame with `pd.DataFrame(rows)`.
- Dictionaries are lookup tables: `series.map(dict)` is a fast way to recode a column.

## Quick quiz

6 questions · tap an answer to check it

**1. What does `d.get("x", 0)` do when "x" is missing?**

A Raises KeyError

B Returns 0

C Adds the key

D Returns None

**Answer: B.** `get` never raises; it returns the default.

**2. Which is a valid dictionary key?**

A `[1, 2]`

B `(1, 2)`

C `{1, 2}`

D `{"a": 1}`

**Answer: B.** Keys must be hashable (immutable).

**3.** Which tool counts occurrences most directly?

A `defaultdict(list)`

B `collections.Counter`

C `set`

D `zip`

**Answer: B.** `Counter(items).most_common()`.

**4.** What does `{"a": 1} | {"a": 2, "b": 3}` give?

A {"a": 1, "b": 3}

B {"a": 2, "b": 3}

C Error

D {"a": 3}

**Answer: B.** The right-hand dictionary wins on conflicts.

**5.** How do you loop over keys and values together?

A `for k in d.values()`

B `for k, v in d.items()`

C `for k, v in d`

D `for v in d.keys()`

**Answer: B.** `items()` yields (key, value) pairs.

**6.** What does `pd.DataFrame(list_of_dicts)` produce?

- **A** A Series
- **B** A table with one row per dictionary and columns from the keys
- **C** A list
- **D** An error

**Answer: B.** This is the standard way to turn JSON-like records into a table.

EXERCISE **Dictionary problems**

(1) Count the frequency of each character in `"banana"`. (2) Group Superstore product `category` into a dictionary of sets of `sub_category`. (3) Invert `{"a": 1, "b": 2}`. (4) Given a list of orders, compute the total per customer using `defaultdict`. (5) Merge two dictionaries where the second overrides the first.

**Show solution (worked answer)**

Python

```python
from collections import Counter, defaultdict
print(Counter("banana"))
g = defaultdict(set)
for c, sc in zip(s["category"], s["sub_category"]): g[c].add(sc)
print({k: sorted(v) for k, v in g.items()})
print({v: k for k, v in {"a": 1, "b": 2}.items()})
orders = [("C1", 100), ("C2", 50), ("C1", 25)]
t = defaultdict(int)
for cid, amt in orders: t[cid] += amt
print(dict(t), {"x": 1} | {"x": 9, "y": 2})
```

INTERVIEW CORNER **Questions you may be asked**

**Q: How does a dictionary achieve O(1) lookup?** It hashes the key to find a slot in a hash table; collisions are resolved by probing, so the average cost is constant.

**Q: `d[k]` vs `d.get(k)`?** `d[k]` raises KeyError for a missing key; `get` returns None (or a default).

**Q: Are dictionaries ordered?** Yes, they preserve insertion order from Python 3.7 onwards.

**Q: Count word frequencies in a text.** `Counter(text.lower().split()).most_common(10)`.

## Practise more online

Free, trusted places to test yourself on this topic. Links open the original site in a new tab.